Modeling California Sea Otter Population

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1 Identifying the Problem

1.1 Background

In California, sea otters are considered a threatened species. They are protected as a result, and we were given the liberty to analyze the otter population for the sake of conservation. We took a look at the California sea otter population to model population growth over a number of years. We used a starting point of 3,000 otters and had a carrying capacity of 16,000 sea otters. The population of otters is affected by health, habitat, and environmental conditions. The growth rates we used to study this population included best, average, and worst intrinsic annual rates. This data comes from the Monterey Bay Aquarium and a UCSB report studying sea otter populations.

1.2 Project Objective

The objective of this project is to compare exponential and logistic models of the sea otter population. A carrying capacity of 16,000 was introduced to keep the population estimates in check for logistic models. We then took a look at 4 ways we can add otters artificially outside of the growth rate to get an idea of unforeseen factors. Finally, we used fixed points to make sure these predictions were in line with realistic goals, ensuring that we stabilized around the carrying capacity.

2 Variables Parameters and Assumptions

2.1 Variables and Parameters

Table 1

Symbol	Meaning	Value or units
n	Number of years since the beginning of the simulation	years
P_n	Sea otter population at year n	otters
P_0	Initial sea otter population	3,000 otters
r	Intrinsic annual growth rate	per year
K	Environmental carrying capacity	16,000 otters
$A(P_n)$	Population added by a management strategy	otters per year

Environmental Growth Rates

- Best conditions: r = 0.052
- Average or medium conditions: r = 0.011
- Worst conditions: r = -0.029

2.2 Modeling Assumptions

Most of our assumptions are pretty standard in order to make our model steady. We assumed that time advanced in one-year steps, the initial population is 3,000 otters in every test, each growth rate remains constant year to year, and the carrying capacity remains 16,000 throughout all tests. Additionally, otters are not separated by age, sex, health, or reproductive status, and random events such as disease, extreme weather, or migration were not considered. Added otters survive and enter the population, and population values may be treated as continuous. Decimal outputs represent continuous estimates, not literal fractions of an otter.

3 Question 1.1 Exponential Population Model

3.1 Developing the Equation

Under exponential growth, the annual change is proportional to the current population. The discrete recurrence is

$$P_{n+1} = P_n + rP_n = (1 + r)P_n$$

Repeated application of the recurrence gives the solution

$$P_n = P_0(1 + r)^n$$

For the three environmental conditions, the recurrences become

$$Best: P_{n+1} = 1.052P_n$$

$$Average: P_{n+1} = 1.011P_n$$

$$Worst: P_{n+1} = 0.971P_n$$

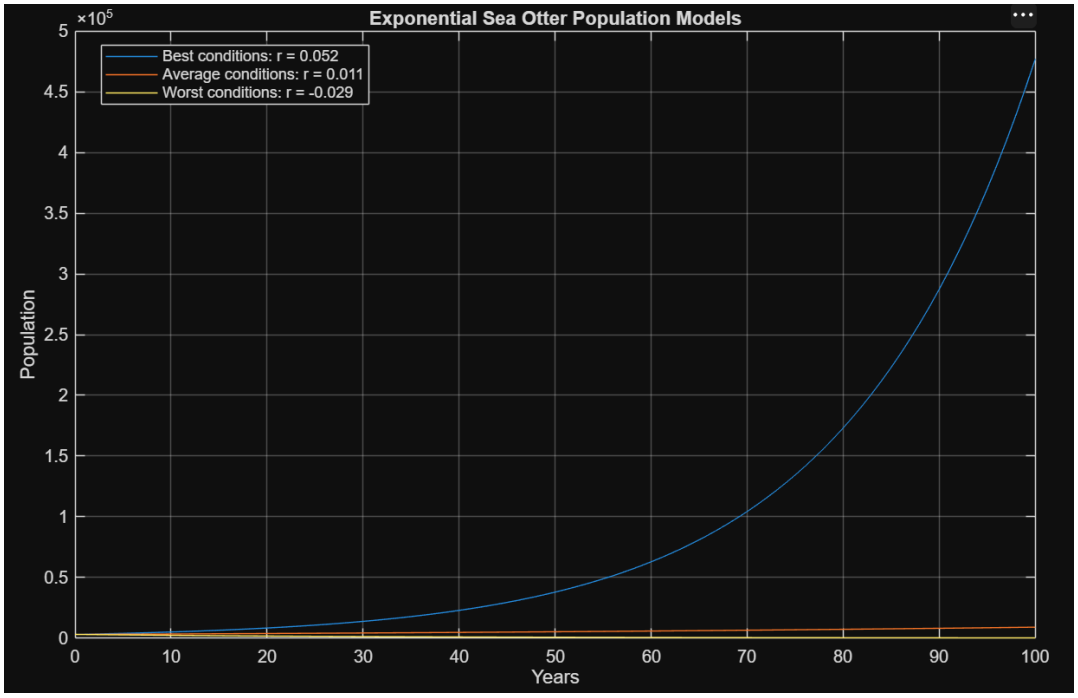
A positive r produces growth, a negative r produces decline, and a larger magnitude of r produces faster change.

3.2 Simulation and Plots

Our average case exponential recurrence model:

average_pop_rate = 3000*(1.011)^years;

Figure 1



In Figure 1, we can see the three growth rates for the best, average, and worst case under an exponential model. After 100 years, we see that the best case resulted in a significantly larger population than both the average and worst rates.

3.3 Interpretation of Results

In Figure 1, we can see that the best and average growth rates increased the population, while the worst case decreased the population by quite a bit. After 10 years, the best case had 4,980.57 otters, the average case had 3,346.82 otters, and the worst case had 2,235.19 otters. After 25 years, the best case had 10,654.10 otters, the average case had 3,943.67 otters, and the worst case had 1,437.48 otters. After 100 years, the best case had 477,194 otters, the average case had 8,958.53 otters, and the worst case had only 158.141 otters. From these results, we can see that the best case pulled away from the other two rates very quickly, while the worst case decreased slightly more quickly than the average increase in population. This is evident in the numbers that show how the best and worst cases pulled away from each other, and even more so from the average.

For this exponential model, we don't have the carrying capacity limiting the otter population. That is why we see nearly 500,000 otters in the 100th year for the best case. It's also why we see all three rates start around the same before they each separate into their different end populations, exploding up or decreasing semi-rapidly. Logically, it makes sense that more animals would produce more animals annually than a population with a smaller number of animals. This exposes a flaw with the exponential growth model: not only would the population likely not grow exponentially in the real world, but there is no limit to what this population ends up as.

4 Question 1.2 Logistic Population Model

4.1 Logistic Equation Development

The logistic model modifies exponential growth by multiplying the growth term by a limiting factor:

$$P_{n+1} = P_n + rP_n(1 - P_n/K)$$

With the assigned carrying capacity, this becomes

$$P_{n+1} = P_n + rP_n(1 - P_n/16,000)$$

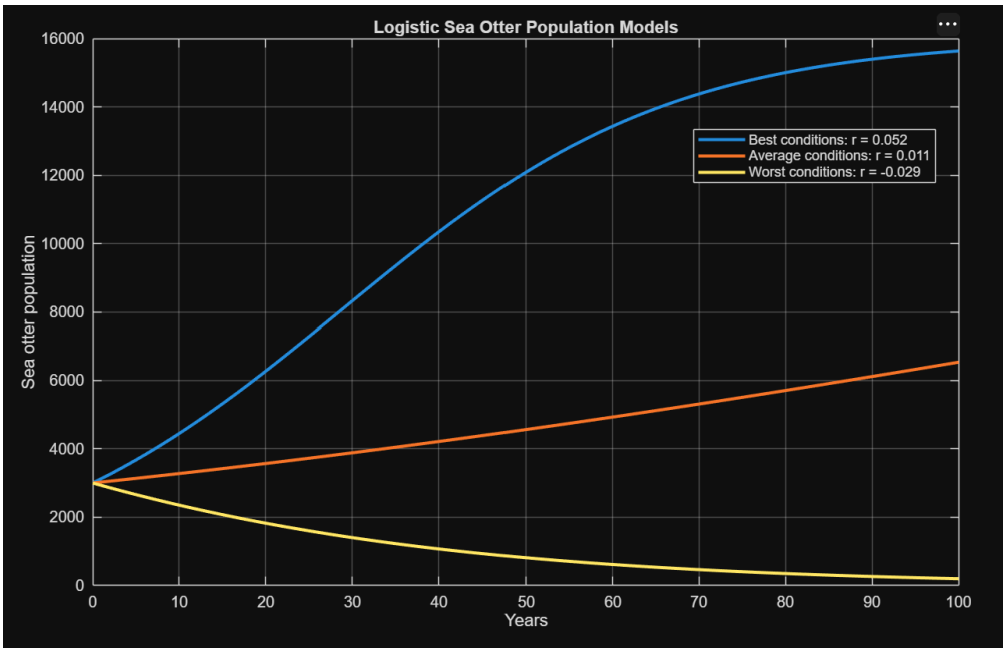
When P_n is much smaller than K, growth is exponential. As P_n approaches K, the limiting factor approaches 0 and growth slows down. At P_n equals K, the modeled natural population change is 0, meaning no change happens. Above K, the logistic growth term becomes negative.

4.2 Simulation and Plots

Our average case logistic recurrence model:

X(j,n) = X(j,n-1) + r(j)*X(j,n-1)*(1 - X(j,n-1)/K);

Figure 2



Sea otter population modeled using a logistic population model, with best, average, and worst cases. We see the best case approach the carrying capacity after 100 years, the average case steadily increases, and the worst case heads towards extinction.

4.3 Interpretation of Results

For Figure 2, we see a much more realistic timeline when compared in Figure 1. It doesn't make sense that so many otters would be added to the population, but the logistic model takes into account more

realistic factors that the exponential model avoids. Additionally, most animals don't see an exponential rate of growth for their populations.

The carrying capacity limits the plot to 16,000, and no lines cross the top of the plot. As is shown in Figure 2, the average case is steadily increasing and should eventually reach the carrying capacity in many, many years. We also see the best case cap out around 16,000, and it may hit it if we were to simulate another 100 years. In the worst case, when r is negative, we see that the population declines to 0, or towards extinction.

The carrying capacity does not prevent decline because it is a ceiling, and not a floor. In the worst conditions, it has no control over whether the population increases or not until it gets close to the capacity. That is why it does not prevent decline.

5 Question 1.3 Adding to the Population

5.1 Developing the Managed Model

Using the average growth rate r = 0.011 and K = 16,000, add a management term $A(P_n)$ to the logistic recurrence:

$$P_{n+1} = P_n + rP_n(1 - P_n/K) + A(P_n)$$

The proposed strategies are

$$Add\ 3\text{: } A(P_n) = 3$$

$$Add\ 10\text{: } A(P_n) = 10$$

$$Add\ 1\ percent\text{: } A(P_n) = 0.01P_n$$

$$Add\ 5\ percent\text{: } A(P_n) = 0.05P_n$$

It is important to note the 2 types of additions here. We see that for 2 plots, we add the same number of otters annually, while for the other 2 plots the amount of otters we add is proportional to the previous year. That means that we should anticipate seeing a stronger long term effect for all of these compared to Figure 2, but more so for the proportional models.

5.2 Simulation and Comparison

Our recurrence model that increases population by an additional 3 otters annually, and is the same for all constants: X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + 3;

Our recurrence model that increases population by an additional 1% annually, and is the same for all percentage additions: X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + X(n-1)*0.01;

Figure 3

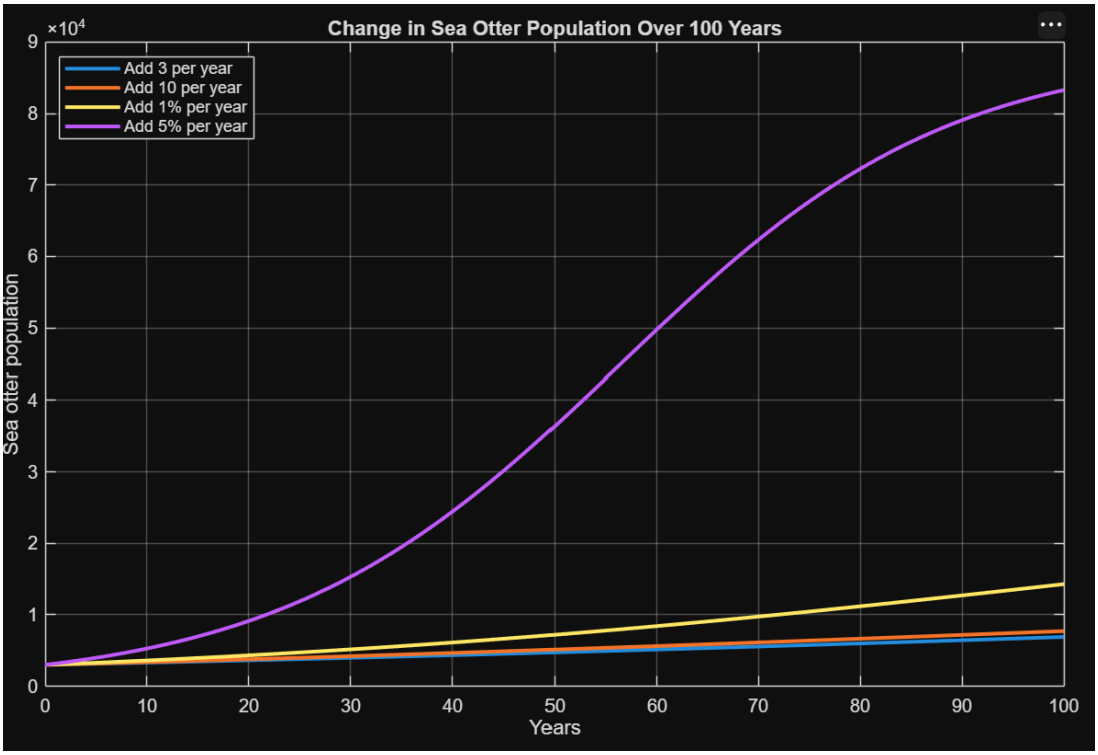


Figure 3 plots all 4 models with additions. It shows the gap between percentage growth and constant growth when added to the natural growth.

Figure 4

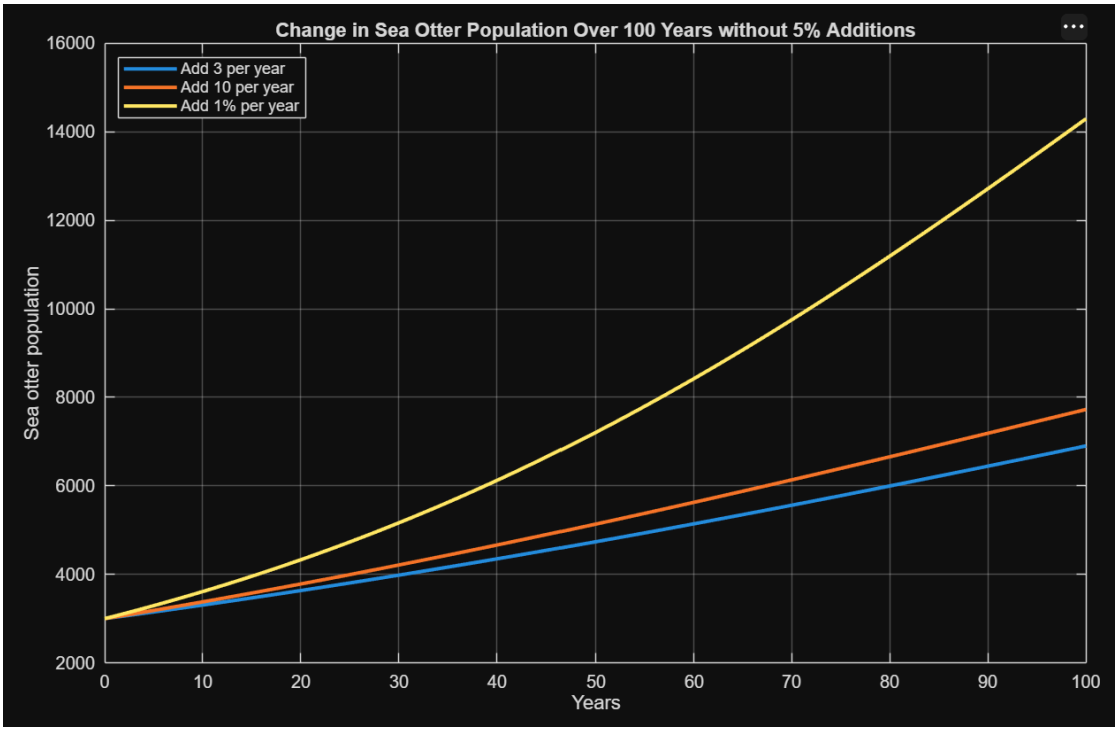


Figure 4 is the same as Figure 3, but without the 5% annual addition. Since the 5% model was so much larger, these values became difficult to analyze.

Table 2

Strategy	P_{10}	P_{25}	P_{50}	P_{100}	Apparent Equilibrium
No additions	3,276.43	3,725.81	4,566.04	6,541	16,000
Add 3	3,307.34	3,806.67	4,737.93	6,903.79	16,268
Add 10	3,379.44	3,995.08	5,136.46	7,727.19	16,863
Add 1 percent	3,612.54	4,732.9	7,209.03	14,296	30,545
Add 5 percent	5,286.98	11,890.4	36,390.3	83,297	88,727

5.3 Equilibria and Interpretation

A nonzero equilibrium P* satisfies

$$P^* = P^* + rP^*(1 - P^*/K) + A(P^*)$$

These new equilibria were taken after running the figure for 1000 years and seeing where stabilization occurred. Normally, the equilibrium will be equivalent to the carrying capacity.

From Figures 3 and 4, we see that only the 5% addition is fast enough for us to see stabilization on the plots. The rest of them are still gradually increasing, but we know when they should begin to stabilize because of the formula. What is interesting and very important to note for all of these plots is the carrying capacity, and how it changes when we add otters outside of the natural growth rate. Under normal circumstances, it would approach 16,000 and stabilize, but when we add otters through unnatural circumstances, the equilibrium actually increases beyond the carrying capacity. For the 2 constant additions, it should still be around 16,000, but we can see for the 5% that the equilibrium will actually be almost 88,000 otters. This just means that the greater our addition, the higher the new equilibrium will be.

The extended simulations provide the equilibrium estimates shown in Table 2. The four strategies have equilibria above the original carrying capacity, while the model with no additions has an equilibrium equal to K. While mathematically valid, it doesn't make a lot of sense ecologically for this to happen. There isn't going to be a random drop of 1% more otters added to the population annually, and that likely still wouldn't make the environment suddenly suitable for more otters than the original carrying capacity.

6 Question 1.4 Stabilization at Carrying Capacity

6.1 Fixed Point Requirement

Write the recurrence as P□₊₁ = f(P□). A fixed point P* satisfies

$$P^* = f(P^*)$$

To stabilize specifically at carrying capacity, the management model must satisfy

$$f(K) = K$$

This is stronger than seeing a simulated curve becoming horizontal because we are saying that the function of the carrying capacity must equal the carrying capacity, ensuring that we stabilize there.

6.2 Proposed Management Strategy

One possible state-dependent supplementation rule is

$$A(P_n) = a\,max(K - P_n, 0)$$

where a controls the fraction of the population deficit addressed each year. The resulting recurrence is

$$P_{n+1} = P_n + rP_n(1 - P_n/K) + a\,max(K - P_n, 0)$$

Because A(K) = 0, substitution gives

$$f(K) = K + rK(1 - K/K) + 0 = K$$

This is just one strategy, but it makes a lot of sense when we just look at it from a real life perspective. If the carrying capacity is 16,000, this reduces the amount of otters being added annually. That way, we can have very large additions of otters, such as 10% of the previous year's deficit annually, that will reach but not exceed the carrying capacity. Then once we reach the capacity, we won't add any more.

To verify mathematically that carrying capacity is a fixed point, we define the recurrence function as

$$f(P) = P + rP(1 - P/K) + a\,max(K - P, 0)$$

A fixed point is a population value that remains unchanged after one recurrence loop. Therefore, a value P* is a fixed point when f(P*) = P*. Substituting the carrying capacity P = K into the function gives

$$f(K) = K + rK(1 - K/K) + a\,max(K - K, 0)$$

Since K/K = 1, resulting term equals zero. The population deficit K - K also equals zero, so the management term equals zero. As a result,

$$f(K) = K + 0 + 0 = K$$

K = 16,000 is a fixed point of the managed model. If the population reaches 16,000 otters, the model predicts no change and no additional otters are added. The population then stays at the carrying capacity.

6.3 Stability Analysis and Plot

This is our recurrence model with the carrying capacity enforced. It uses variables to make it easier to understand:

X(j,n) = previousPopulation + naturalChange + addedOtters;

Showing that K is a fixed point proves that a population already at K will remain there. For a population below carrying capacity, max(K - P, 0) is equivalent to K - P. The recurrence function can be written as:

$$f(P) = P + rP(1 - P/K) + a(K - P)$$

Because the max function changes P = K, the full function is not differentiable at K when a > 0. Since our simulations begin below K and approach it, we use this formula:

$$f'(P) = 1 + r - 2rP/K - a$$

Evaluating the derivative gives us:

$$f'(K) = 1 - r - a$$

A fixed point is stable when |f′(K)| < 1. When we plug the average case, r = 0.011, into this equation, we get:

$$f'(K) = 0.989 - a$$

For the tested values a = 0.01, a = 0.05, and a = 0.10, the corresponding derivative values are 0.979, 0.939, and 0.889. Each absolute value is less than one, so K is a stable fixed point for each strategy.

Figure 5

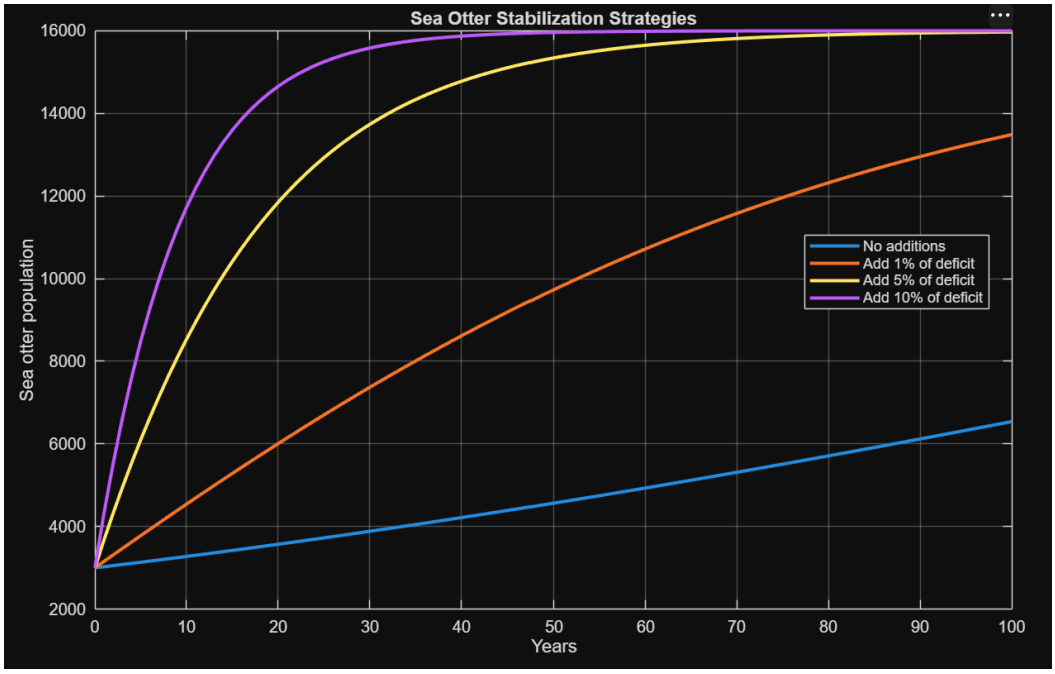


Figure 5 compares the unmanaged model with strategies that add 1%, 5%, and 10% of the population deficit annually.

6.4 Interpretation of Results

This strategy preserves K, 16,000, as a fixed point because we treat it as a hard cap. Once we get close to it, as we can see in the 5% and 10% additions, it will smooth out and never cross K. That is because our equation takes into account when we reach the carrying capacity, and stops adding otters to it annually. Larger values of the variable a move the population towards K faster than the unmanaged logistic model. Unlike the additions in Section 5, the deficit-based addition decreases as P_n approaches K, so K stays as the equilibrium.

The result becomes more sensitive to our chosen value of a when it reaches the carrying capacity too quickly. For 5% and 10%, we see a sharp curve to stabilize at 16,000. For 1%, however, we see a smooth bowing curve as it gets closer. Once the curves stabilize, and really after they finish bowing, we can expect the annual rate of otters added to decrease. As the population increases, the remaining population deficit decreases, so the number of otters added decreases every year.

This strategy seems pretty natural, ecologically realistic and ethically correct. I would imagine that most populations have something like this happen where once a certain population is reached, the additions begin to slow down. We can assume that the death rate increases in these cases from environmental factors, and that explains why otters are added at a lesser rate once a threshold is approached. As mentioned earlier, when we didn't have a way to stop adding, the constant and percentage based additions shifted the equilibrium above K. We see now what happens when we enforce that cap instead of letting the annual addition dictate what the new cap will be.

7 Conclusion

We analyzed how different environmental growth rates affected population behavior over a century. We saw how the exponential model isn't a great representation of this problem because the numbers become much too large, and exponential growth isn't really how population growth is represented logically in the real world. Once we added the carrying capacity, it prevented these super large populations and modeled it in a more realistic way. Still, logistic growth models lack randomness in their equations, so it is hard to say that this was the best way to model population growth.

We added fixed and proportional amounts of otters to the logistic model. Once we added these sub-populations, we encountered a new problem: they far exceeded our equilibrium beyond our original carrying capacity. This was due to the equation not accounting for the new additions, resulting in the sub-population creating a new equilibrium for the model. We fixed this by using the remaining deficit as a budget, decreasing the number of otters added once the population enters the range of the carrying capacity, thus keeping the equilibrium equal to the carrying capacity.

This strategy ended up being the most appropriate logically, mathematically, and ecologically. Logically, it makes sense that once we near the cap, we stop adding otters to the population. Mathematically, the equation only added a max function to implement this. The equation otherwise remained equivalent to the other models we tested. Ecologically, this also makes sense if we already take into account the carrying capacity. If the environment did not provide enough space for the number of otters, then it would make sense that the carrying capacity acts as a hard ceiling, and adding more would only result in migration or death of other otters from causes like food scarcity.

7.1 Limitations and Possible Extensions

The most plausible limitation is that growth rates will vary over time. Our tests assumed they would stay the same, but that is rarely true in real life, especially over a century. Other limitations could include changing carrying capacity depending on habitat conditions, random environmental events, or migration of subpopulations. These all could have a huge impact on our findings and all tests we did since they would add a factor that doesn't currently exist to our equations. They would all be more realistic than our current tests, so they are still interesting and relevant for future research.

8 References

Sea Otter, Monterey Bay Aquarium Animals A-to-Z. https://www.montereybayaquarium.org/animals-the-ocean/animals-a-to-z/sea-otter

Population Dynamics and Biology of the California Sea Otter (Enhydra lutris nereis) at the Southern End of its Range. 2006 Coastal Research Center, Marine Science Institute, University of California, Santa Barbara.

Appendix A Code

```
% 1.1 !!!
figure(1)
years = 0:100; % integer years
pop1 = 3000*(1.052).^years;% population each year
pop2 = 3000*(1.011).^years;
pop3 = 3000*(0.971).^years;
plot(years, pop1, years, pop2, years, pop3)
xlabel('Years')
ylabel('Population')
legend('Best conditions: r = 0.052', ...
'Average conditions: r = 0.011', ...
'Worst conditions: r = -0.029', ...
'Location', 'best')
grid on
title('Exponential Sea Otter Population Models')

% 1.2 !!!
figure(2)
K = 16000;
X0 = 3000;
r = [0.052, 0.011, -0.029];
years = 0:100;
X = zeros(3, length(years));
X(:,1) = X0;
for j = 1:3
          for n = 2:length(years)
                    X(j,n) = X(j,n-1) + r(j)*X(j,n-1)*(1 - X(j,n-1)/K);
          end
end
plot(years, X(1,:), 'LineWidth', 2)
hold on
plot(years, X(2,:), 'LineWidth', 2)
plot(years, X(3,:), 'LineWidth', 2)
xlabel('Years')
ylabel('Sea otter population')
title('Logistic Sea Otter Population Models')
legend('Best conditions: r = 0.052', ...
'Average conditions: r = 0.011', ...
'Worst conditions: r = -0.029', ...
'Location', 'best')
grid on

% 1.3 !!!
figure(3)
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(1, length(years));
X(1,1) = X0;
for n = 2:length(years)
          X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + 3;
end
plot(years, X(1,:), 'LineWidth', 2)
title('Sea Otter Population with 3 Otters Added Annually')
xlabel('Years')
ylabel('Sea otter population')
grid on

figure(4)
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(1, length(years));
X(1,1) = X0;
for n = 2:length(years)
X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + 10;
end
plot(years, X(1,:), 'LineWidth', 2)
title('Sea Otter Population with 10 Otters Added Annually')
xlabel('Years')
ylabel('Sea otter population')
grid on
```

```
figure(5)
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(1, length(years));
X(1,1) = X0;
for n = 2:length(years)
          X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + X(n-1)*0.01;
end
plot(years, X(1,:), 'LineWidth', 2)
title('Sea Otter Population with 1% Annual Growth')
xlabel('Years')
ylabel('Sea otter population')
grid on

figure(6)
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(1, length(years));
X(1,1) = X0;
for n = 2:length(years)
          X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + X(n-1)*0.05;
end
plot(years, X(1,:), 'LineWidth', 2)
title('Sea Otter Population with 5% Annual Growth')
xlabel('Years')
ylabel('Sea otter population')
grid on

figure(7)
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(4, length(years));
X(:,1) = X0;
for n = 2:length(years)
          previous = X(:,n-1);
          additions = [3; 10; 0.01 * previous(3); 0.05 * previous(4)];
          X(:,n) = previous + r .* previous .* (1 - previous/K) + additions;
end
plot(years, X(1,:), 'LineWidth', 2)
hold on
plot(years, X(2,:), 'LineWidth', 2)
plot(years, X(3,:), 'LineWidth', 2)
          legend('Add 3 per year', …
          'Add 10 per year', …
          'Add 1% per year', …
          'Location', 'northwest')
title('Change in Sea Otter Population Over 100 Years without 5% Additions')
xlabel('Years')
ylabel('Sea otter population')
grid on

figure(8)

K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(4, length(years));
X(:,1) = X0;
for n = 2:length(years)
          previous = X(:,n-1);
          additions = [3; 10; 0.01 * previous(3); 0.05 * previous(4)];
          X(:,n) = previous + r .* previous .* (1 - previous/K) + additions;
end
plot(years, X(1,:), 'LineWidth', 2)
hold on
plot(years, X(2,:), 'LineWidth', 2)
plot(years, X(3,:), 'LineWidth', 2)
plot(years, X(4,:), 'LineWidth', 2)
legend('Add 3 per year', ...
```

```
'Add 10 per year', ...
'Add 1% per year', ...
'Add 5% per year', ...
'Location', 'northwest')
title('Change in Sea Otter Population Over 100 Years')
xlabel('Years')
ylabel('Sea otter population')
grid on

% 1.4 !!!
figure(9)
clf
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
% Test 0, 1%, 5%, and 10% additions
a = [0, 0.01, 0.05, 0.10];
X = zeros(length(a), length(years));
X(:,1) = X0;
for j = 1:length(a)
          for n = 2:length(years)
                    previousPopulation = X(j,n-1);
                    naturalChange = r*previousPopulation * (1 - previousPopulation/K);
                    populationDeficit = max(K - previousPopulation, 0);
                    addedOtters = a(j)*populationDeficit;
                    X(j,n) = previousPopulation + naturalChange + addedOtters;
          end
end
plot(years, X(1,:), 'LineWidth', 2)
hold on
plot(years, X(2,:), 'LineWidth', 2)
plot(years, X(3,:), 'LineWidth', 2)
plot(years, X(4,:), 'LineWidth', 2)
hold off
title('Sea Otter Stabilization Strategies')
xlabel('Time (Years)')
ylabel('Sea otter population')
legend('No additions', …
          'Add 1% of deficit', …
          'Add 5% of deficit', …
          'Add 10% of deficit', …
          'Location', 'best');
grid on
xlim([0 100])
```

A.1 Exponential model code

```
years = 0:100; % integer years
pop1 = 3000*(1.052).^years;% population each year
pop2 = 3000*(1.011).^years;
pop3 = 3000*(0.971).^years;
```

A.2 Logistic model code

```
K = 16000;
X0 = 3000;
r = [0.052, 0.011, -0.029];
years = 0:100;
X = zeros(3, length(years));
X(:,1) = X0;
for j = 1:3
     for n = 2:length(years)
               X(j,n) = X(j,n-1) + r(j)*X(j,n-1)*(1 - X(j,n-1)/K);
     end
end
```

A.3 Population-addition strategy code

```
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(1, length(years));
X(1,1) = X0;
```

```
for n = 2:length(years)
     X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + 3;
end
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(1, length(years));
X(1,1) = X0;
for n = 2:length(years)
     X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + 10;
end
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(1, length(years));
X(1,1) = X0;
for n = 2:length(years)
     X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + X(n-1)*0.01;
end
K = 16000;
X0 = 3000;
r = 0.011;
years = 0:100;
X = zeros(1, length(years));
X(1,1) = X0;
for n = 2:length(years)
     X(1,n) = (X(1,n-1) + r*X(1,n-1)*(1 - X(1,n-1)/K)) + X(n-1)*0.05;
end
```

A.4 Stabilization strategy code

```
for j = 1:length(a)
     for n = 2:length(years)
               previousPopulation = X(j,n-1);
               naturalChange = r*previousPopulation * (1 - previousPopulation/K);
               populationDeficit = max(K - previousPopulation, 0);
               addedOtters = a(j)*populationDeficit;
               X(j,n) = previousPopulation + naturalChange + addedOtters;
     end
end
```